

# ART IN MOTION

## A GESTURE-BASED INTERACTIVE DRAWING SYSTEM



- Drawing through hand gestures, no mouse or keyboard
- Real-time interaction using computer vision
- Designed to support children's creativity
- Playful, intuitive, and sustainable digital art

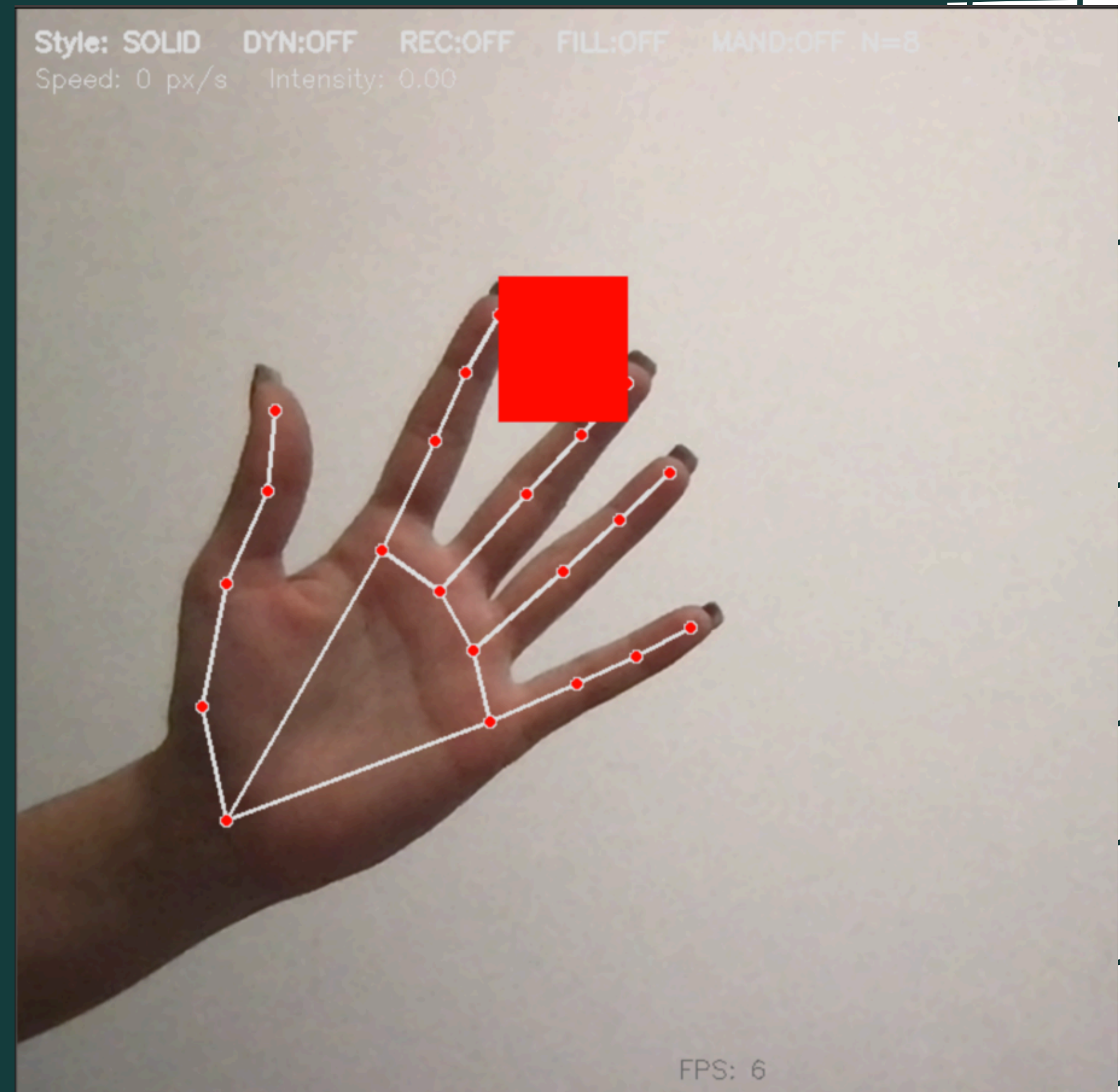
# **SYSTEM ARCHiTECTURE**

- **Real-time loop in Python (OpenCV)**
- **Hand tracking with MediaPipe**
- **Gesture interpretation**
- **Drawing engine**
- **Canvas blending**

# GESTURE-BASED

## iNTERACTiON

- Selection mode (index + middle)
- Drawing mode (index)
- Pinch control (thumb + index)
- Landmark-based finger states
- Mode switching
- Pinch distance control
- Temporal smoothing



## EXPORT

**SALVA PNG**  
imagine finale

## MODES

DYN ON

DYN OFF

REC ON

REC OFF

FILL ON

FILL OFF

## MANDALA

MAND ON

MAND OFF

SECT -  
N=8

SECT +  
N=8

## STYLE

SOLID

DOTTED

DASH

SPRAY

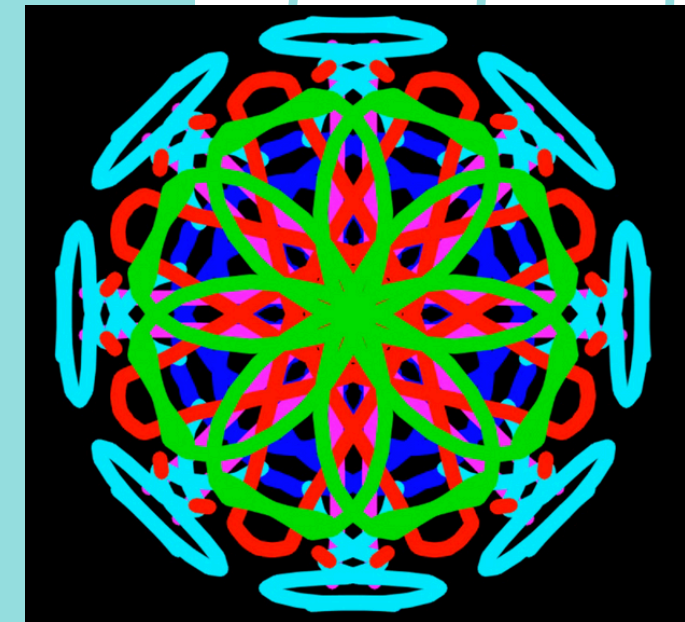
MARKER

NEON

CHALK

QUIT

# DRAWING ENGINE



- Persistent canvas + mask
- Multiple stroke styles
- Dynamic color mapping
- Mandala symmetry
- Media export

# SOURCES & TECHNOLOGIES

## Core stack

- Python
- OpenCV (cv2) – real-time capture, rendering, recording
- MediaPipe Hands – landmark-based gesture control
- NumPy – canvas and mask management

## Execution

- Real-time, CPU-only
- 1280×720, 30 FPS

# **FUTURE WORK & EXTENSIONS**

## **Interaction**

- **Multi-hand / multi-user drawing**
- **Depth-based hand tracking**

## **Platform & performance**

- **Tablet / AR versions**
- **GPU acceleration**

## **Interface & evaluation**

- **Advanced UI frameworks**
- **User studies in real contexts**